

Bilingual Document Q&A (RAG) — System & Evaluation Report

Repository: OmarKhalil2003/bi-RAG | Corpus: 24 Public Docs (12 AR, 12 EN) | Architecture: Hybrid (BGE-M3 + BM25 RRF)

1. Approach & Design Rationale

This project implements a lightweight, production-ready Bilingual (Arabic & English) Document Question Answering (RAG) system with measured retrieval and generation quality. The design emphasizes simplicity, zero framework bloat, and strict factual grounding.

- **Corpus & Chunking:** 24 public documents (12 Arabic, 12 English) covering government policies, technical standards (NIST, ISO, ECC-1:2018), and water/environmental reports. Text is extracted via PyMuPDF with Unicode NFKC normalization and diacritic cleaning. Documents are split into 500-token chunks with 75-token overlap using paragraph-aware boundaries, prepended with contextual headers ([Document: {title} | Page: {page}]) to preserve provenance.
- **Baseline Retrieval (Dense):** BAAI/bge-m3 embeds queries and passages into 1024-dimensional normalized vectors stored in a FAISS IndexFlatIP database for exact cosine search.
- **Enhanced Retrieval (The Single Improvement — Hybrid Search):** To eliminate neural reranker latency while ensuring precise matching for regulatory codes and numeric thresholds, we integrated sparse lexical retrieval via BM25Okapi. Dense and sparse candidate lists are fused using Reciprocal Rank Fusion (RRF, k=60):

$$Score_RRF(doc) = 1 / (60 + Rank_dense(doc)) + 1 / (60 + Rank_bm25(doc))$$

- **Generation & Dual Refusal Gates:** The generation layer matches the query language (Arabic or English), cites supporting passages with [source: chunk_id], and refuses unanswerable queries. A retrieval gate directly refuses queries when dense score < 0.50 and BM25 score < 15, preventing LLM hallucinations before generation.

2. Quantitative Evaluation

Evaluated on a fixed gold benchmark of 40 questions (16 Arabic answerable, 16 English answerable, 4 Arabic unanswerable, 4 English unanswerable) assessing factual lookups, numerical thresholds, and cross-lingual queries.

Table 1: Retrieval Performance Comparison (Hit@k & MRR)

Architecture	Hit@1	Hit@3	Hit@5	MRR
Baseline (Dense BGE-M3 + FAISS)	1.0000	1.0000	1.0000	1.0000
Enhanced Hybrid (Dense + BM25 RRF)	0.9688	0.9688	0.9688	0.9688

Finding: Arabic queries scored 1.0000 across both systems. On English query q_en_16 (cross-lingual query targeting an Arabic policy), dense search ranked the true passage at Rank 1, while BM25 suffered zero lexical overlap, demonstrating the standard cross-lingual trade-off of unweighted hybrid search.

Table 2: Generation Quality & Refusal Performance

Metric	Result	Evaluated Subset	Description
Answer Correctness	68.75% (22/32)	Answerable queries	Exact verified facts and citations
Partial Correctness	18.75% (6/32)	Answerable queries	Core facts correct; minor clause omission
Combined Correctness	87.50% (28/32)	Answerable queries	Factually grounded answers
Unsupported-Answer Rate	3.12% (1/32)	Answerable queries	Unevidenced numbers/claims (1 query)
Unanswerable Refusal	100.0% (8/8)	Unanswerable queries	Deterministic refusal without hallucination

3. Failure Analysis (10 Representative Cases)

We conducted a detailed analysis of all 10 non-perfect cases on the gold benchmark to pinpoint failure modes:

#	Query ID & Topic	Observed Failure Mode	Proposed Fix
1	q_ar_02 (Cybersecurity MFA)	Dense score high (0.719), but low BM25 score triggered false refusal	Bypass refusal when dense confidence $\geq$ 0.60
2	q_ar_03 (AI Ethics Principles)	Numbered list digits ('4.', '5.') flagged as unsupported numbers	Strip list numbering prefixes before fact verification
3	q_ar_05 (Air Quality PM2.5)	PM2.5 answered; secondary electric bus target from page 2 missed	Aggregate adjacent page chunks for multi-target queries
4	q_ar_10 (Water Strategy)	Desalination target answered; leakage reduction timeline missed	Sub-query decomposition for multi-part questions
5	q_ar_15 (Distribution Network)	100% coverage cited; 2-hour pipe repair timeframe missed	Completeness checklist prompt verifying all question clauses
6	q_en_04 (OECD AI Principles)	All 5 principles listed; 'high-risk' classification trigger omitted	Enforce concise bullet answers without conversational preamble
7	q_en_08 (Cloud HSM BYOK)	FIPS 140-2 Level 3 cited; annual key rotation schedule omitted	Instruct model to extract operational lifecycles alongside ratings
8	q_en_09 (GDPR Data Breach)	72h window cited; fine clause phrasing scored partial	Accept flexible phrasing for complex statutory penalties
9	q_en_14 (CISA Incident Playbook)	SHA-256 verification cited; minor forensic synonyms scored partial	Add standard technical synonyms to gold evaluation reference
10	q_en_16 (Cross-Lingual Cloud)	Zero English-Arabic BM25 overlap promoted irrelevant English docs	Disable BM25 when query and document languages diverge

4. Service Implementation & Prioritized Fixes

- **FastAPI Service:** Exposes POST /query (validates query input, performs hybrid retrieval, returns answer with citations, refusal status, and latency) and GET /health (checks FAISS index, BM25 store, and embedding models). Complete unit test suite (13 tests) validates chunking, retrieval metrics, and API contracts.
- **Prioritized Fixes:** (1) Dynamic BM25 Weighting: deactivate BM25 on cross-lingual queries to prevent lexical noise; (2) Sub-Query Decomposition: decompose multi-part questions into individual lookups; (3) On-Premise Small LLM: package an offline model (e.g. Qwen2.5-3B) to guarantee zero API dependency.